

# **EduNexus: A Web Based Application for School Management System**

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## **FINAL YEAR DESIGN PROJECT REPORT**

This Report Presented in Partial Fulfillment of the Requirements for  
the **Degree of Bachelor of Science in Computer Science and  
Engineering**

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# APPROVAL

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This Project titled “**EduNexus: A Web Based Application for School Management System**”, submitted by **Sumaiya Amrin**, ID No: **221-15-5040** and **Md. Sakib Hossain**, ID No: **221-15-4677** to the Department of Computer Science and Engineering, Daffodil International University has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.S.C in Computer Science and Engineering and approved as to its style and contents. The presentation was held on **09 December, 2025**.

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# DECLARATION

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We hereby declare that this project has been done by us under the supervision of **Mr. Mohammad Jahangir Alam, Assistant Professor**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere for the award of any degree or diploma.

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# ABSTRACT

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This project introduces EduNexus, a complete web-based School Management System that unifies academic and administrative tasks into one platform. It covers student information management, class and subject assignment, attendance tracking, result submission, and administrative analytics. It is designed to replace the manual, paper-based processes still used in many schools in Bangladesh. Many schools in Bangladesh still depend on manual record-keeping, which leads to errors, delays, mismanaged attendance, and a lack of real-time academic insights. EduNexus solves these issues by offering a centralized and efficient solution that reduces manual workload, increases accuracy, enhances transparency, and improves institutional productivity. Additionally, the system allows room for future expansion, such as advanced reporting dashboards, mobile support, automated analytics, and integration with digital learning tools. EduNexus centralizes a specific school operation by allowing administrators to manage classes, subjects, teachers, and students in one system, while teachers can take attendance, view class-wise student lists, and submit subject-wise results efficiently. The application is developed using React, JavaScript-ES6, HTML, and Tailwind CSS for the frontend and Node.js, Express.js, and REST APIs for the backend to ensure a seamless, fast, and scalable user experience. It also implements role-based access control, where Admins have full data-management privileges and Teachers can access only class- and subject-related features assigned to them.

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# Chapter 1

## Introduction

The title of our project that this chapter introduces is "***EduNexus: A Web-Based Application for School Management System.***" This chapter starts with the background and the problem statement, followed by the motivation for choosing this topic. The purpose is to explain why such a system is needed and how it is valuable to solve real-life problems in school management.

### 1.1 Introduction

The name of this project is "EduNexus: Web-based School Management System Application." The application has been developed with React for the front-end and Node.js with Express for the backend, using REST APIs that ensure seamless communication across system parts. EduNexas is designed to simplify day-to-day school functions by delivering faster, simpler, and more accurate digital workflows across both academic and administrative activities. Instead of traditional handwritten records or fragmented manual processes, this system provides an end-to-end solution for class, subject, teacher, student, attendance, and result management. Most of the schools in Bangladesh are still manually managed, which increases repeated problems like misplacing student information, incorrect tracking of attendance, inconsistency in result entry, and delays in accessing academic performance data of students [2], [9], [19]. These problems increase further with the incremental increase in the number of students, teachers, and subjects every year. EduNexus reduces these challenges by providing class and subject management, teacher assignments, handling student information, recording attendance, and class-wise result submissions. Furthermore, the system also provides yearly admission statistics, failure records, and analytics of daily attendance to the administrators for making data-driven decisions in the institution. The system has secure role-based access wherein Admins will have full CRUD on subjects, teachers, students, and classes, while Teachers have permission only to view assigned class details, take attendance, and submit results. This split ensures accuracy, control, and transparency regarding school workflows. In the end, EduNexas enhances operational efficiency and reduces administrative workload, thus offering a state-of-the-art digital platform tailored to the needs of educational institutes.

### 1.2 Motivation

The main motivation behind developing ***EduNexus*** is to make school management more organized, efficient, and transparent by utilizing modern digital technology. In many schools across Bangladesh, important academic and administrative records are still maintained manually on paper or in scattered notebooks [2], [9], [19]. This not only consumes extra time but also leads to frequent errors in student information handling, attendance tracking, class-wise data management, and result processing. These problems become even more

challenging when schools grow in size and the volume of data increases. We wanted to solve this real-world issue by building a system that allows one to store all school-related information properly, securely access it, and manage it easily through a single centralized digital platform.

### 1.3 Objectives

The primary aim of this project is to develop an easy-to-use School Management System to minimize the workload of administrators and teachers. The overall efficiency of daily academic and administrative activities is enhanced as well. It digitizes major school activities, such as student information management, class and subject setup, teacher assignment, attendance record keeping, and result submission. The system automates these crucial tasks, making school operations faster, reliable, and not as prone to human error.

In brief, the objectives of *EduNexus* include:

- Minimize or completely eliminate the use of manual paper-based record keeping in the schools.
- It minimizes errors in managing student information, class data, attendance, and academic results [2], [3].
- Provide teachers and admins with an easy, quick, and safe means to manage daily school tasks.
- Making the system user-friendly in a way that even admins and teachers will be able to operate it without technical difficulties.

### 1.4 Methodology

In developing EduNexas, a structured, step-by-step methodology has been followed to ensure that the system meets the needs of both the school administrators and the teachers, while operating efficiently and accurately. The major steps were:

1. **Problem Identification:** Researched existing manual management methodologies of schools; identified the following issues: difficult maintenance of student records, inconsistent tracking of attendance, delayed processing of academic results, and non-centralized data management.
2. **Requirement Collection:** Determine the essential functional requirements such as student management, class and subject setup, teacher assignment, attendance recording, and result submission. Requirements for role-based access for Admin and Teacher users were also defined.
3. **System Design:** Designed the overall system architecture, database schema, and user flow. Defined user roles, module interactions, and class–subject–teacher mapping. Prepared layout structures for the Admin Dashboard and Teacher Dashboard.
4. **Implementation:** The system was implemented using React, JavaScript (ES6), HTML, Tailwind CSS for the frontend, and Node.js with Express for the backend. REST APIs were developed to securely and efficiently communicate data between all modules.

5. **Testing and Validation:** Examined all system functionalities by using sample datasets to generate accurate attendance calculations, result submissions free of errors, and smooth user interactions secured under correct role-based access control. Finalization: Refined the UI, optimized system performance, and integrated all modules for smooth operation by Admin and Teacher users.

## 1.5 Project Outcome

The major deliverable of this project is a fully functional web-based application for schools to manage their daily academic and administrative activities in an orderly and efficient manner. Basically, **EduNexas** digitalizes the process of keeping students' records, class and subject management, teacher assignment, attendance recording, and academic results submission. School administrators and teachers would save lots of manual work and also avoid common mistakes which happen in paper-based systems after using this.

Expected outcomes of the project are:

- A user-friendly application that is operable by admins and teachers without advanced technical knowledge.
- Precise student management and class management: all updates regarding the student record, class assignment, and subject are stored instantly and securely.
- Simple and efficient recording of attendance by teachers; hence, daily tracking is quicker and more accurate.
- Easy academic result submission for accurate class-wise and subject-wise performance records.
- Secure role-based access, where Admin and Teacher permissions are properly maintained to protect sensitive school data.
- Centralized system dashboard for admins to view yearly admissions, daily attendance summaries, and yearly failure statistics.
- Improved data management that ensures significant academic and administrative information is not lost or misplaced.

## 1.6 Organization of the Report

The report is divided into six chapters to ensure clarity, organization, and that the project will be clearly presented.

### • Chapter 1: Introduction

This chapter provides the background of the project. It includes the problem statement, the motivation for the system, the project's objectives, a summary of the methodology, expected outcomes, and an overview of the report structure.

### • Chapter 2: Background

This chapter discusses the background study relevant to digital school management systems. This encompasses the literature review, an examination of existing school information systems, alternative applications, and a gap analysis that guided the development of the proposed solution.

### • Chapter 3: Research Methodology

The methodology that has been used to develop EduNexas is detailed in this chapter. It describes requirement analysis, which includes both functional and non-functional requirements, system design, data flow diagrams, user interface design, and overall project planning and task allocation.

- **Chapter 4: Implementation and Results**

This chapter describes the implementation details of the system using selected technologies. It covers the development environment, module-wise implementation, testing procedures, performance evaluation, and the results obtained during the testing phase.

- **Chapter 5: Engineering Standards and Design Challenges**

This chapter talks about the engineering standards followed throughout the project, problems in design and implementation, and how the system adheres to software, hardware, and communication standards. This section further examines the impacts of the system on society, the environment, and sustainability, along with financial considerations and connections to complex engineering problems.

- **Chapter 6: Conclusion**

This chapter summarizes the results of the project, points out the limitations, and discusses some future improvements that could be made to enhance the system.

# Chapter 2

## Background

This chapter provides the background details needed by the project. It includes an introduction of the area, a review of existing works, and investigation of related systems. The purpose is to equip readers with enough information so they can understand the rest of the report.

### 2.1 Introduction

Application of digital technologies in educational institutions has become vital in recent years. Large volumes of student information, attendance, academic results, and administration are handled by schools regularly. In most of the schools in Bangladesh, these activities are still performed using paper-based systems and manual records. The process is well-known and easy to handle, but it often leads to errors when maintaining student information, delay in accessing academic results, and the risk of misplacing important documents [2].

In view of these problems, Computerized School Management Information Systems have been introduced in most parts of the world. Such systems are able to keep student and academic data securely, automate attendance, manage class and subject assignments, handle result submissions, and provide speedy access to vital information about the school. They save time, reduce paperwork, improve accuracy, and reduce human errors in daily academic and administrative tasks [3], [4], [6].

Our EduNexas project has integrated these benefits into a complete and functional web-based school management solution. Leveraging the power of modern technologies such as React for the frontend and Node.js with Express for the backend, EduNexas will be in a position to provide schools with a reliable and cost-effective platform to make their operations seamless. The background research upon which this project stands supports how digital school management systems benefit from removing the restrictions placed by the traditional manual workflow, thereby enriching the efficiency of school administration and academic management in general [15], [16].

## 2.2 Literature Review

**Table 2.1:** Summary of Literature Reviewed.

| Author (s)          | Year | Title                                               | Methodology                          | Key Findings                                                                                                      |
|---------------------|------|-----------------------------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Olaniyi et al. [1]  | 2016 | Development of a Web-Based School Management System | System development & case analysis   | Found that digital school systems reduce errors in student record handling and improve administrative efficiency. |
| Adewale & Ayo [2]   | 2016 | ICT-Driven Education Management Information System  | Framework analysis                   | Showed that EMIS enhances data accuracy and supports decision-making in schools.                                  |
| Nkosi & Moyo [3]    | 2018 | Evaluation of School Information Systems            | Survey-based research                | Concluded that SMIS improves workflow but requires proper user training for effectiveness.                        |
| Awuah [4]           | 2019 | Effects of SMIS on Administrative Performance       | Observational & statistical analysis | Reported improved administrative productivity and better institutional planning.                                  |
| Hussain & Javed [5] | 2019 | Cloud-Based School Information System               | Prototype testing                    | Developed a cloud-based SMIS that enhanced accessibility and data security.                                       |
| Boateng [6]         | 2019 | Role of ICT in Improving School Management          | Case study                           | Found that ICT adoption reduces manual workload and increases operational accuracy.                               |
| Singh & Kumar [7]   | 2017 | Web-Based Student Attendance & Management System    | System design                        | Demonstrated that automated attendance systems reduce time and human errors.                                      |
| Mwalongo [8]        | 2016 | Teachers' Perceptions of School Information Systems | Qualitative study                    | Identified usability and training as major factors influencing system adoption.                                   |

| Author (s)          | Year | Title                                  | Methodology      | Key Findings                                                                      |
|---------------------|------|----------------------------------------|------------------|-----------------------------------------------------------------------------------|
| Rashid & Rahman [9] | 2020 | ICT Integration in Bangladeshi Schools | Survey research  | Highlighted common issues in manual school processes and the need for automation. |
| UNESCO [10]         | 2018 | EMIS Standards & Best Practices        | Policy framework | Emphasized the global need for standardized digital school management practices.  |

### 2.2.1 Similar Applications

**Table 2.2:** Similar Applications.

| Application                                         | Location                              | Key Features (Already on Service)                                                                                                                                                              |
|-----------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Development of a Web-Based School Management System | Global (used in Asia, Europe, Africa) | Open-source school management system; student information system, attendance, gradebook, timetable, messaging, role-based access, parent portal. Popular among small-medium schools worldwide. |
| openSIS – OS4ED                                     | USA, Bangladesh, India                | Cloud-based SIS; student demographics, attendance, discipline, scheduling, report cards, transcripts, health records. Used by thousands of schools globally.                                   |
| EduHive – Smart Education ERP                       | Bangladesh                            | All-in-one school ERP with attendance, online classes, fees, exam results, LMS integration, SMS alerts. Widely deployed in Bangladeshi schools.                                                |
| PRAN-RFL School ERP (Internal Use)                  | Bangladesh                            | Custom-built ERP for institutional schools; class management, teacher profiles, attendance system, reports, and controlled dashboard access. Used in multiple PRAN-run schools.                |
| School Management System – CodeClouds               | India, Bangladesh                     | Web-based management solution; student admission, attendance, online fees, homework, exam management, teacher modules, and progress reports. Implemented in various South Asian schools.       |



## 2.3 Gap Analysis

A review of existing school management systems reveals that although many applications are already in service, they frequently come short of delivering important features like flexible role-based access, class-subject-teacher mapping, simplification of attendance handling, and integrated result management dashboards [1], [3], [8], [9]. EduNexas was designed to fill these gaps by integrating the essential features both in academics and administration into a single system.

**Table 2.3:** Gap Analysis of Existing Systems vs. EduNexas

| Features                                                    | Gibbon | openSIS | EduHive (BD) | School ERP (India) | CodeClouds SMS | Proposed system(EduNexas) |
|-------------------------------------------------------------|--------|---------|--------------|--------------------|----------------|---------------------------|
| Student Information Management                              | Yes    | Yes     | Yes          | Yes                | Yes            | Yes                       |
| Class-Subject-Teacher Mapping                               | No     | Yes     | Yes          | Yes                | Yes            | Yes                       |
| Role-Based Access (Admin, Teacher)                          | Yes    | Yes     | Yes          | Yes                | Yes            | Yes                       |
| Teacher Assignment for Multi-Class Subjects                 | No     | No      | Yes          | Yes                | No             | Yes                       |
| Attendance Management                                       | Yes    | Yes     | Yes          | Yes                | Yes            | Yes                       |
| Result Submission by Subject Teacher                        | No     | Yes     | Yes          | Yes                | No             | Yes                       |
| Admin Dashboard Analytics (admission, attendance, failures) | No     | No      | No           | Yes                | No             | Yes                       |
| Student Result View (Class Teacher)                         | No     | Yes     | Yes          | Yes                | Yes            | Yes                       |
| Modern Tech Stack (React, Node, REST API)                   | No     | No      | No           | No                 | No             | Yes                       |
| Multi-branch / Multi-school Support                         | Yes    | Yes     | Yes          | Yes                | Yes            | Yes                       |
| Ease of Use for Non-Technical Users                         | No     | No      | Yes          | Yes                | No             | Yes                       |
| Affordable for Small/Medium Schools                         | Yes    | No      | Yes          | No                 | Yes            | Yes                       |

## 2.4 Summary

This chapter presents the background study that justifies the development of our project, **EduNexus: A Web-Based School Management System Application**. First, it clarified the imperative of adopting computerized systems in school operations and how such systems will drastically reduce the problems associated with manual record-keeping practices. Secondly, from the literature review, we searched for various related studies which proved that computerized SMIS increases administrative efficiency, reduces human errors, and improves institutional management performance.

We also reviewed a number of existing school management applications, both local and international, that are currently in use. Their strengths and weaknesses were analyzed to understand where they succeed and where they fail. Following this, a detailed gap analysis was performed comparing our proposed solution with other available systems. The results indicated that while many existing solutions offer student information management, attendance tracking, and reporting, they often lack integrated class–subject–teacher mapping, strict role-based access for teachers and admins, and a seamless system designed specifically for Bangladeshi school operational needs.

This chapter outlined the requirements that are missing in current systems and emphasized the need for EduNexus to be developed. The following chapter discusses the methodology, study of requirements, and approach towards system design followed to successfully complete the project.

# Chapter 3

## Research Methodology

This chapter presents the methodology followed in the development of EduNexus, dealing with the analysis of the requirements, system design, and specifications that guided the implementation of the project. This chapter thus attempts to present how the project was planned, organized, and structured to achieve accuracy, reliability, and usability within a real school environment. In a systematic manner, this chapter outlines how the different steps in requirements gathering, system component designs, and implementation of the functionalities have been carried out so as to shape up EduNexus into an integrated and effective school management solution.

### 3.1 Requirement Analysis & Design Specification

#### Requirement Analysis

First, we identified the requirements of the school management system. We studied the daily operations of schools in Bangladesh and their common challenges: inaccuracy in manual attendance tracking, inability to handle large volumes of student records, disorganization in class-subject-teacher management, delays in result processing, and lack of adequate administrative reporting. Based on these observations, the **functional requirements** of **EduNexus** were set as follows:

- Add, update and delete details about students, teachers, subjects, and classes.
- Assign class and subject teachers to numerous classes and subjects.
- Allow teachers to take attendance only for the classes they supervise.
- Permit the teachers to submit results only for the subjects assigned to them.
- Provide admins with dashboard analytics such as admissions, attendance summary, and yearly failure statistics.
- Implement role-based access for Admin and Teacher securely, with restricted permissions.
- Store and manage class-wise lists of subjects and the assigning of teachers for the same.
- Class-wise results can be viewed by the class teachers themselves.

The **non-functional requirements** include:

- **Usability:** The system should be simple and easy to use for both admins and teachers.
- **Security:** Sensitive school data should be protected with secure authentication and role-based permission control.
- **Performance:** The application will respond promptly for all attendance, result submission, and class data operations.
- **Scalability:** The system must support future expansion, such as multi-school or multi-branch management.

These non-functional requirements include usability, security, performance, and scalability, which are essential quality attributes for dependable and sustainable educational software systems [13], [21].

## Design Specification

We then designed the system architecture and workflows after the analysis of the requirements. EduNexas was designed as a full web based system with the technologies as follows:

- **Frontend:** React, JavaScript (ES6), HTML5, and Tailwind CSS of responsive and user-friendly interfaces.
- **Backend:** API development, server-side logic and role-based validation: Node.js with Express.
- **Database:** MongoDB on which to store the student, teacher, class, subject, attendance, and result data safely.

We have created use case diagrams, data flow diagrams (DFD) and entity-relationship (ER) models in order to describe system functionality in a very clear way. The architecture was designed to have a hierarchical role-based access model in which:

- Admins can do all CRUD on students, classes, and teachers, and subjects.
- The teachers can only access information on their assigned classes, attendance, and results of their assigned subjects.

Every workflow had been developed to provide the user with a smooth experience and enable successful functioning of all school management processes.

### 3.1.1 Overview

EduNexas is a school management system that is developed as an online application that facilitates the organization of the information of students, the structure of the classes, the assignment of the teachers, the attendance and also the academic outcomes. The primary objectives of the system are to substitute the manual and paper-based systems with a digital system, which is not only quick and accurate but also simple to navigate.

The system has a client server setup. The frontend is written in React that allows creating a responsive, interactive, user-friendly interface to both admins and teachers. The backend is developed on Node.js and Express that support business logic, secure processing of API and authentication and role-based access control. The database is deployed in MongoDB in which all the crucial data, including student profiles, teacher records, classes, subjects, attendance records, and results are stored safely.

The system contains two categories of users:

- The whole school platform is handled by the Admin. It involves the addition and upgrading of students, teachers, classes, and subjects, allocation of class teachers and subject teachers, and reviewing of analytics like annual admissions, attendance reports, and failure rates.
- The teacher undertakes academic activities, which include watching classes assigned, attending daily classes, subject-wise results, and a summary of results class-wise in case of being a class teacher.

The system has the following key functions: Subject management in classes, the correct attendance monitoring system, a secure role-based system, organized results submission, and automated dashboard analytics. The architecture should be secure, easy to use and scale, and meet the needs of the small and medium-sized schools now and be easy to upgrade in the future with multi-branch school management and sophisticated analytics.

### 3.1.2 System Design

The EduNexas system design is the entire architecture of the school management application and the way various phases of development were implemented. It is designed according to

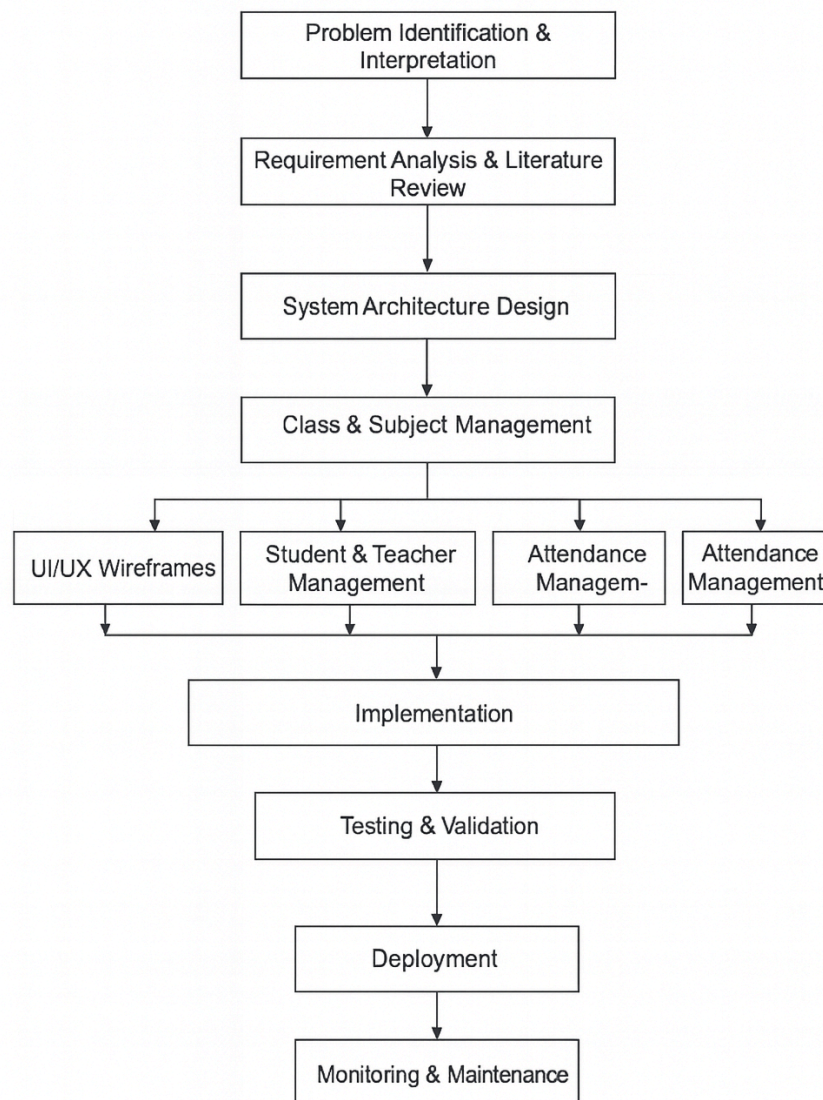
the Waterfall Model that is a sequential development model because every stage has to be fulfilled before proceeding to the next stage. The reason why this model was selected is due to the fact that it represents a clear structured workflow, enables detailed documentation, and can be used in the course of academic projects with clear and stable requirements.

The system flow starts with the identification of the problem and requirement analysis, then the system architecture and database schema design is designed. Once the core structure has been established the next step in the project is detailed module design, which is detailed UI/UX layouts, student and teacher management, class-subject mapping, attendance processing, and role-based access control. After the design has been developed, the development is done with React as the front-end and Node.js and Express as the back-end and MongoDB as the database. The system is then subjected to rigorous testing and validation to make it accurate and usable and the final stage is deployment to actual operation use. Monitoring and maintenance is the last process and ensures that the system will be reliable in the long run and it can be extended with new functions as required.

EduNexas is based on the following modules:

- **UI/UX Wireframes** - To design clean, intuitive, user-friendly interfaces of the Admin and Teacher dashboards.
- **Student and Teacher Management** - To add, update, assign and keep extensive records of students and Teachers.
- **Class and Subject Management** - To manage classes, subjects and relate teachers with certain subjects in various classes.
- **Attendance Management** - To be able to enter attendance to classes that are assigned to them on a daily basis.
- **Result Submission Module** - To enable subject teachers to post marks and the class teachers to access class-wise results.
- **Role Management** - To implement secure and role-based access in the role of an Admin and Teacher to ensure that every user is allowed to do only what he or she should.

EduNexas architecture is expected to be secure, scalable, and maintainable, which implies the unproblematic functioning of schools of different sizes and makes it possible to expand the architecture without significant structural alterations in the long run.



**Figure 3.1:** EduNexas System Design (Waterfall Model)

The above figure (Figure 3.1) depicts how the design of the EduNexus system would be according to the Waterfall Model [11], [12]. It demonstrates the evolution of the project step by step, which starts with problem identification and requirement analysis, which is followed by the system architecture design, class and subject structuring, and finally the module-level development. Once the design stage has been fulfilled, implementation of the system is done using the chosen front-end and backend technologies and comprehensive testing and validation of the system is undertaken to validate the correctness and reliability.

This is a linear and structured way of development, where each stage must be finished first before an individual moves on to the next stage and this makes the whole process well documented, systematic and more convenient to maintain. The Waterfall Model also promotes straightforward planning, task traceability, and orderly development, thus is very appropriate to academic software projects like EduNexus.

### 3.1.3 Functional and Nonfunctional Requirements

#### Functional Requirements

The functional requirements explain the tasks that EduNexus will be required to do to facilitate proper running of schools. The following are the key features that enable the administrators, teachers, students, and parents to do their respective duties in the system:

- **Student Management:** create, edit, delete and view student records containing information about students like personal information, class, section, guardians information, and records. Filter the class wise students also.
- **Teacher Management:** Teacher management, including teacher profiles, subject assignments, workload, as well as attendance and performance records.
- **Class Management:** Establish and maintain classes, sections and subject distributions in a manner that has the right schedule.
- **Attendance Management:** Teachers will be able to record daily attendance of students and administrators can check attendance summaries.
- **Result and Grade Management:** The teachers are able to post up marks, create results and automatically compute grades according to the grading system of the institution.
- **User Role Management:** Each user should have separate user roles, allow Admin, Teachers each role should have distinct permissions and its dashboard.
- **Communication System:** Admin should view the teacher activities and students performances. Teachers can login by the admin access permissions. By doing these teacher and admin communication will be built up.
- **Report Generation:** Show attendance, student performance, teacher workload, and class summaries reports.
- **Authentication and security:** authenticate all users by use of username & passwords, handle sessions and validate input.

#### Non-Functional Requirements

Non-functional requirements outline how EduNexas is supposed to work in order to make the system efficient, usable, and secure:

- **Usability:** The system has to be user friendly to users with a low level of technical knowledge, in which it has to be easy to use and have an intuitive interface.
- **Performance:** Other activities like submission of attendance, loading of reports and fees should be fast without futile delays.
- **Scalability:** The system must be capable of serving more students, teachers, classes, as well as branches in future.
- **Security:** Secure authentication, validation, encrypted messages, and access must be used to protect all data.
- **Reliability:** It is expected to be a stable system with no failures, and more importantly, no failure in the operations that are essential to the system like attendance and fee management.
- **Maintainability:** The system architecture must facilitate the easy revision, debugging and incorporation of new modules whenever necessary.
- **Portability:** EduNexas, being a web-based application, should work on various kinds of devices (desktop, laptop, tablet, mobile) and on the representatives of major web browsers.

### 3.1.3 Data Flow Diagram

The flow of information in the School Management System is demonstrated in the chapter with the help of Data Flow Diagrams (DFDs). DFDs break down the system into the highest level to the logical subprocesses, identify external entities, processes, data stores, and flows.

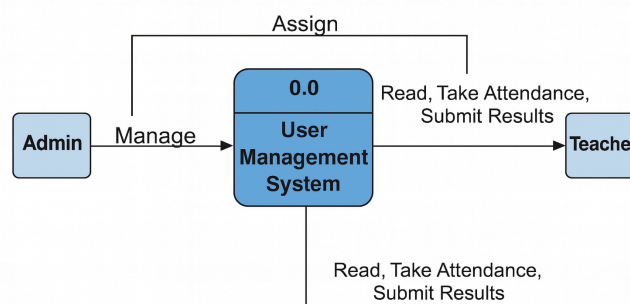
#### Diagram Symbols and Notations

| Name                   | Description                                                                            | Symbol                                                                                |
|------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Process</b>         | Represent transformations of input data into output data.                              |    |
| <b>External Entity</b> | Represent people or systems outside the boundary that interact with the school system. |    |
| <b>Data Store</b>      | Represent repositories where data is stored for later use.                             |  |

**Figure 3.2:** Notation and Symbol of DFD

#### DFD Level-0 (Context Diagram)

The Level-0 DFD, or context diagram, is the way that the whole EduNexus: School Management System is depicted as the one and the only process. It demonstrates the total interaction between the system and the external parties that communicate with the system.



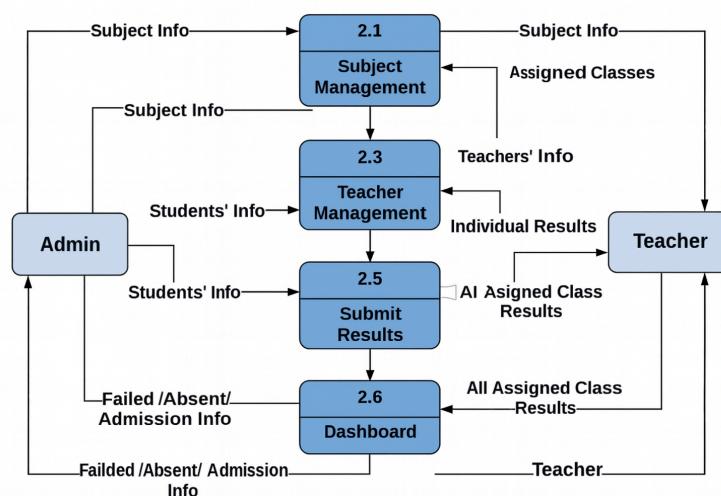
**Figure 3.3:** DFD Level-0 (Context Diagram) – EduNexus



## DFD Level-1 (Major Processes)

Level-1 DFD divides the EduNexus system into four main core processes which are the main operations managed in the school management system. These processes demonstrate the interaction of student data, academic data, administrative processes and reporting functions in the system.

- **Manage Student Information (1.1):** This module takes care of all the student record operations:
  1. Admit students by filling up the admission form with all information.
  2. Renewal of personal data.
  3. Saving of section/roll/class information.
  4. Control of parent or guardian information.
- **Academic information management (1.2):** This process retains academic information pertaining to:
  1. Class assigned with subject plan.
  2. Subjects assigned.
  3. Teacher assignments.
  4. Academic Result performances.
- **Manage Attendance & Results (1.3):** It is among the fundamental modules of the day-to-day school functions. It includes:
  1. Making attendance of students.
  2. Marking/grading of each subject.
  3. Computation of results and final grades.
  4. Revising performance records that are available to class teachers.
- **Generate Reports (1.4):** In this module, all the necessary academic and administrative reports are created:
  1. Attendance summaries.
  2. Student report cards of individual students.
  3. Performance analytics by the class.
  4. Teacher reports.
  5. Administrative briefs on school authorities.



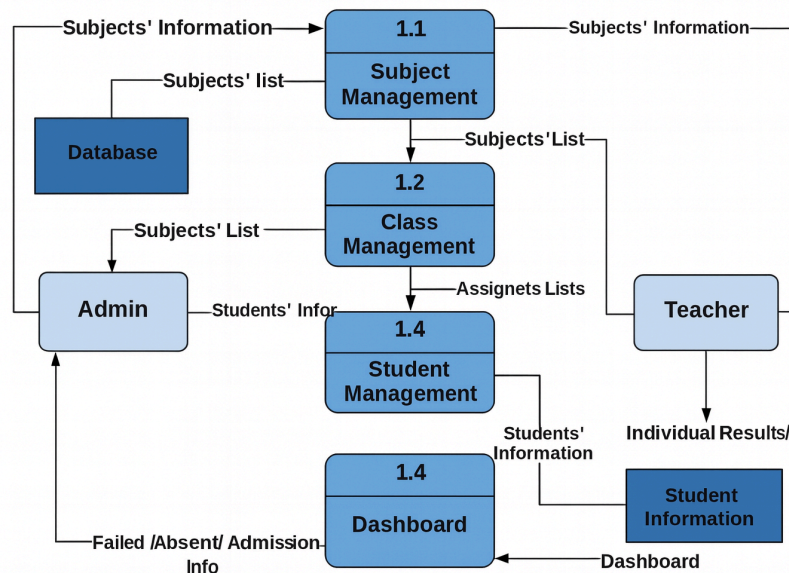
**Figure 3.4:** DFD Level-1 Major Processes of EduNexus

## DFD Level-2 (Detailed Process - Attendance and Result Management)

The Level-2 DFD then decomposes the Manage Attendance and Results (1.3) process into internal processes in detail. These processes demonstrate how attendance and academic performance of students are documented, handled, and stored in the EduNexus.

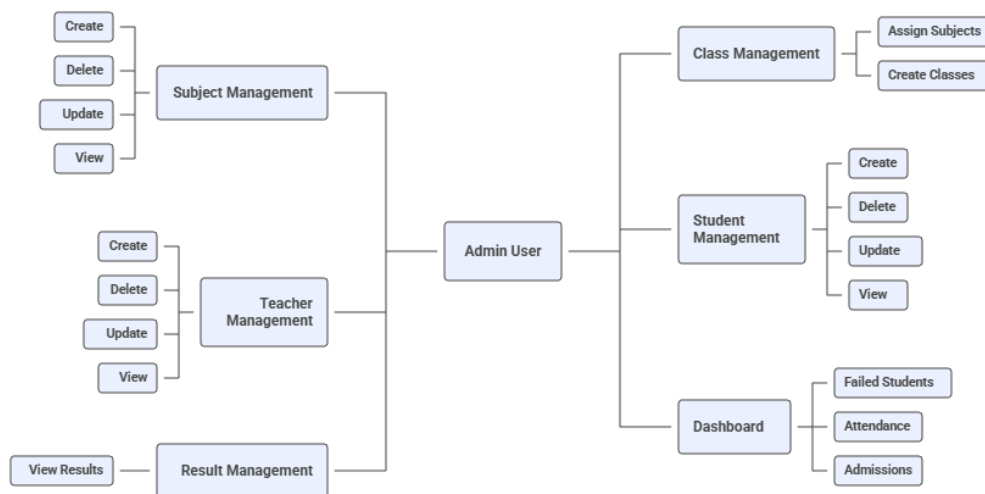
The breakdown is as follows detailing:

- **Select Class/Student:** To initiate the attendance process or result-entry, teachers pick a class, subject and then students list in the system.
- **Registering the attendance / Mark entry:** The students are marked by the teacher with Present, Absent. To write tests, marks per subject are typed in the system. Entries to the system are checked to prevent duplication or improper formatting.
- **Process Review Result / Summarize Process Attendance:** The system automatically: The percentage of attendance calculated on a daily, monthly, and annual basis. Calculates cumulative marks, grades and GPA according to school policies. This is to make sure that there is accuracy and removes errors in manual calculations.
- **Lift Student Records and Compile Report:** The system adds to the centralized database of students: Attendance logs, marks and grade sheets.  
Finally, EduNexus generates:
  1. Attendance reports.
  2. The results sheets of individual students.
  3. Summary of performance in classes.
  4. Admin and class teacher can view them online and print them.



**Figure 3.5:** DFD Level-2 (Detailed Process - Attendance and Result Management)

## Use Case Diagram :



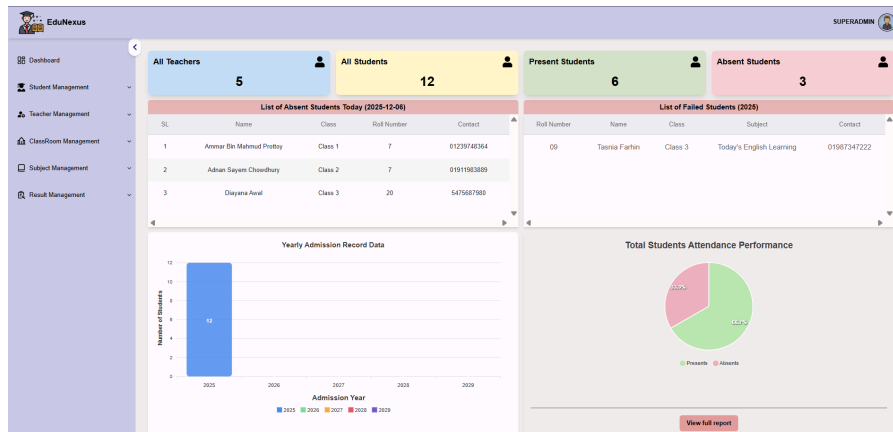
**Figure 3.6:** Use Case Diagram

### 3.1.4 UI Design

The login page features a light blue background. At the top center is an icon of a person wearing a graduation cap and holding an open book, with the text "লগ ইন করুন" (Log In) below it. Below the icon are two input fields: "User Name" and "Password". The "Password" field has a toggle icon (an eye) to its right. Below the "Password" field is a CAPTCHA image showing the characters "zvd5" and a refresh icon to its right. Below the CAPTCHA is a text box with the placeholder "Type the characters above". At the bottom is a dark blue button with the text "LOG IN" in white.

**Figure 3.7:** (Login Page for Multiple Users)

Admin will login with a fixed password , password id given in the backend code and teacher will login by email as username and contact number as password.



**Figure 3.8:** (Admin Dashboard)

Admin can manage students, teachers, classroom and subjects with CRUD operation. Can view the daily attendances of every class's students and can view the results and failed students. And also the admission record.

The Students List view allows administrators to manage individual student records. It includes a sidebar with navigation options and a main section titled "Student's Records" with an "Add Student" button. A "Filter by Class" dropdown is set to "All". The table below lists the students:

| Class   | Name                  | ID | Age | Sex    | Religion | Email                | View Details         | Update               | Delete                 |
|---------|-----------------------|----|-----|--------|----------|----------------------|----------------------|----------------------|------------------------|
| Class 1 | Faiyaz Hossain        | 03 | 7   | Male   | Muslim   | faiyaz23@gmail.com   | <a href="#">View</a> | <a href="#">Edit</a> | <a href="#">Delete</a> |
| Class 1 | Sumaiya Amrin         | 04 | 7   | Female | Muslim   | sumaiya6@gmail.com   | <a href="#">View</a> | <a href="#">Edit</a> | <a href="#">Delete</a> |
| Class 1 | Monika Kabir          | 01 | 7   | Female | Muslim   | monika44@gmail.com   | <a href="#">View</a> | <a href="#">Edit</a> | <a href="#">Delete</a> |
| Class 2 | Ashik Hossain         | 9  | 10  | Male   | Muslim   | ashik90@gmail.com    | <a href="#">View</a> | <a href="#">Edit</a> | <a href="#">Delete</a> |
| Class 2 | Punit Soni            | 27 | 13  | Male   | Hindu    | putn32@gmail.com     | <a href="#">View</a> | <a href="#">Edit</a> | <a href="#">Delete</a> |
| Class 2 | Adnan Sayem Chowdhury | 7  | 13  | Male   | Muslim   | adnan15@edu.cd.com   | <a href="#">View</a> | <a href="#">Edit</a> | <a href="#">Delete</a> |
| Class 3 | Diayana Awal          | 20 | 14  | Female | Muslim   | diayana220@gmail.com | <a href="#">View</a> | <a href="#">Edit</a> | <a href="#">Delete</a> |
| Class 3 | Tasnia Farhin         | 09 | 12  | Female | Muslim   | tasnia09@gmail.com   | <a href="#">View</a> | <a href="#">Edit</a> | <a href="#">Delete</a> |
| Class 4 | Hafizur Rahman Fahim  | 08 | 14  | Male   | Muslim   | hafizur20@gmail.com  | <a href="#">View</a> | <a href="#">Edit</a> | <a href="#">Delete</a> |

**Figure 3.9:** (Students List)

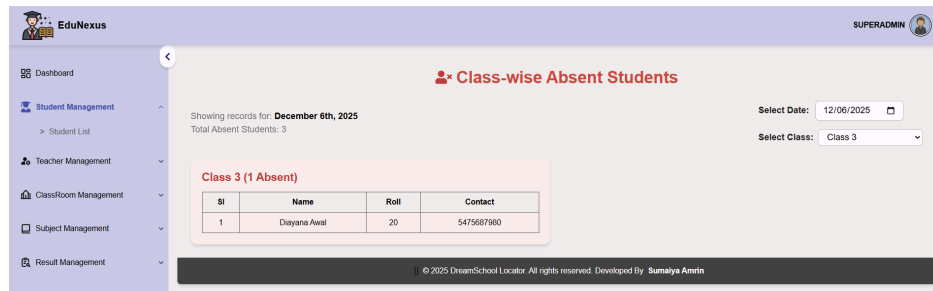
The "Admit New Student" form is used to register new students. It includes a "Back" button and two main sections:

- Student Personal Information:**
  - Student ID (text input)
  - Name (text input)
  - Class (dropdown menu)
  - Age (text input)
  - Sex (dropdown menu)
  - Email (text input)
- Guardian & Others Information:**
  - Father's Name (text input)
  - Mother's Name (text input)
  - Guardian Contact (text input with placeholder +8801XXXXXXX)
  - Guardian Profession (text input)
  - Religion (dropdown menu)
  - Caste (text input)
  - Blood Group (dropdown menu)
  - Date of Birth (date picker)

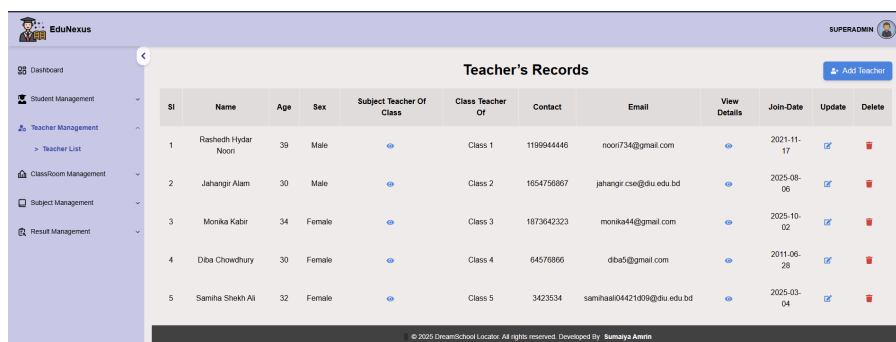
**Figure 3.10:** (Admit Student Form)



**Figure 3.11: (ClassWise Present Students)**



**Figure 3.12:(ClassWise Absent Students)**



**Figure 3.13: (All\_Teachers List)**

**Figure 3.14: (Add\_Teacher Form)**

| Class Name        | Room No | ClassWise Subject Plan                                                                                        | Update | Delete |
|-------------------|---------|---------------------------------------------------------------------------------------------------------------|--------|--------|
| Class 1           | 101     | Today's English Learning, Mathematics & Geometry, Advance English, বাংলা ১ম পত্র                              |        |        |
| Class 2           | 204     | Today's English Learning, Mathematics & Geometry, বাংলা ১ম পত্র                                               |        |        |
| Class 3           | 302     | Today's English Learning, Mathematics & Geometry, বাংলা ১ম পত্র                                               |        |        |
| Class 4           | 401     | Today's English Learning, Mathematics & Geometry, বাংলা ১ম পত্র                                               |        |        |
| Class 5           | 409     | Today's English Learning, Mathematics & Geometry, বাংলা ১ম পত্র, Social Science                               |        |        |
| Class 6(A)        | 601     | বাংলা ২য় পত্র, English For Today, Mathematics & Geometry, English 2nd Paper, বাংলা ১ম পত্র, Social Science   |        |        |
| Class 6(B)        | 602     | English 2nd Paper                                                                                             |        |        |
| Class 7           | 701     | English Literature                                                                                            |        |        |
| Class 8           | 709     | Advance English                                                                                               |        |        |
| Class 9(Science)  | 900     | বাংলা ২য় পত্র, জীববিদ্যা, প্রাণিবিদ্যা, English Literature, পদার্থবিদ্যা, রসায়ন, উচ্চতর গণিত, বাংলা ১ম পত্র |        |        |
| Class 9(Commerce) | 903     | Mathematics & Geometry                                                                                        |        |        |
| Class 10(Science) | 909     | জীববিদ্যা, প্রাণিবিদ্যা, রসায়ন                                                                               |        |        |
| KO_9              | 908     | বাংলা ২য় পত্র, Mathematics & Geometry 4                                                                      |        |        |

**Figure 3.15: (All ClassRoom List)**

Class Wise Subject plans are fetched from the Subject Management.

**Figure 3.16: (Add\_Class's Information)**

| SubjectCode | SubjectName              | Description                                 | Update | Delete |
|-------------|--------------------------|---------------------------------------------|--------|--------|
| Eng-03      | Today's English Learning | Textbook Story, Written & Reading           |        |        |
| Engle-06    | বাংলা ২য় পত্র           | বাংলা বাকসমূহ এবং ভাষা                      |        |        |
| BIO-09      | জীববিদ্যা, প্রাণিবিদ্যা  | উদ্ভিদবিদ্যা এবং জীববিজ্ঞান, প্রাণিবিদ্যা   |        |        |
| BAN-03      | English Literature       | English Written and Reading                 |        |        |
| Math-09     | Mathematics & Geometry   | Learn, Apply & Play                         |        |        |
| ENG-01      | English For Today        | Stories and Poem                            |        |        |
| PHY_10      | পদার্থবিদ্যা             | পদার্থবিদ্যা ১ম পত্র, পদার্থবিদ্যা ২য় পত্র |        |        |
| CHEM-09     | রসায়ন                   | রসায়ন ১ম পত্র এবং ২য় পত্র                 |        |        |
| CHEM-10     | রসায়ন                   | রসায়ন ১ম পত্র এবং ২য় পত্র                 |        |        |
| HM-09       | উচ্চতর গণিত              | উচ্চতর গণিত                                 |        |        |
| HM-10       | উচ্চতর গণিত              | উচ্চতর গণিত                                 |        |        |
| ENG-06      | Advance English          | Test                                        |        |        |
| Math-09     | Mathematics & Geometry   | Math world                                  |        |        |

**Figure 3.17: (SubjectList Management)**

### Add New Subject

← back

Subject Code

Subject Name

Description

Submit Subject

**Figure 3.18:** (Add\_Subjects Form)

The screenshot shows the 'All Student's Result Performance' page. On the left is a sidebar with navigation links: Dashboard, Student Management, Teacher Management, Classroom Management, Subject Management, and Result Management. The 'Result Management' section is expanded, showing 'All Students Result'. The main content area has a 'Select Class' dropdown menu open, displaying a list of classes: Class 1, Class 2, Class 3, Class 4, Class 5, Class 6(A), Class 6(B), Class 7, Class 8, Class 9 (Science), Class 9 (Commerce), Class 10 (Science), and KO\_9.

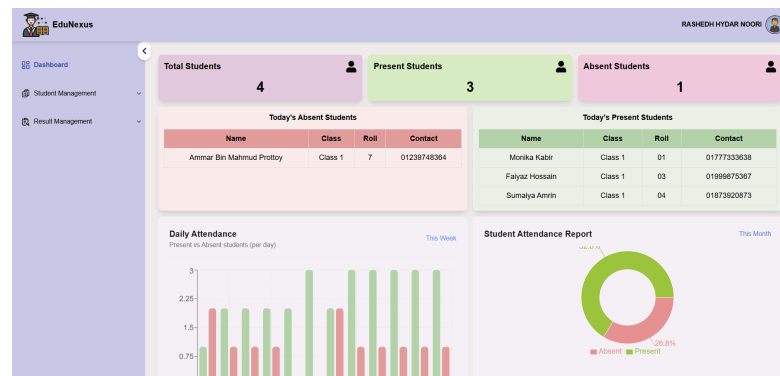
**Figure 3.19:** (AdminResult-[admin can view classwise all students result])

First, to view the specific class's student's result, the admin needs to choose the class from the class list.

The screenshot shows the 'Academic Result Report' page. It displays the student's name (Puntl Sori), ID (27), and class (Class 2). Below this is a table titled 'Result Progress' showing the student's performance in three subjects.

| Subject Name             | Total Marks | Grade |
|--------------------------|-------------|-------|
| Today's English Learning | 85          | A     |
| Mathematics & Geometry   | 92          | A+    |
| গণিত ১৭-৭১৪              | 88          | A     |

**Figure 3.20:** (Example of result of any specific class's any student)



**Figure 3.21: (ClassTeacher's Dashboard)**

Class Teacher can view specific students information, attendees, result view & submission

| Student ID | Name                     | Class   | Age | Sex    | Religion | Email               | Details |
|------------|--------------------------|---------|-----|--------|----------|---------------------|---------|
| 01         | Monika Kabir             | Class 1 | 7   | Female | Muslim   | monika44@gmail.com  | Details |
| 03         | Faiyaz Hossain           | Class 1 | 7   | Male   | Muslim   | fayaz23@gmail.com   | Details |
| 04         | Sumaiya Aminin           | Class 1 | 7   | Female | Muslim   | sumaiya6@gmail.com  | Details |
| 7          | Ammar Bin Mahmud Prottoy | Class 1 | 15  | Male   | Muslim   | prottoy75@gmail.com | Details |

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**Figure 3.22: (Specific Class's Student's List)**

| Student ID | Name                     | Status                                      |
|------------|--------------------------|---------------------------------------------|
| 01         | Monika Kabir             | <input checked="" type="checkbox"/> Present |
| 03         | Faiyaz Hossain           | <input checked="" type="checkbox"/> Present |
| 04         | Sumaiya Aminin           | <input checked="" type="checkbox"/> Present |
| 7          | Ammar Bin Mahmud Prottoy | <input type="checkbox"/> Absent             |

Submit Attendance

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**Figure 3.23:(Attendance Sheet Under Class Teacher)**

| Student ID | Name                  | Status                                      |
|------------|-----------------------|---------------------------------------------|
| 7          | Adnan Sayem Chowdhury | <input type="checkbox"/> Absent             |
| 9          |                       | <input checked="" type="checkbox"/> Present |
| 27         |                       | <input checked="" type="checkbox"/> Present |

Attendance Submitted Successfully For Today

Attendance submitted successfully!

**Figure 3.24:(Attendance's Submission UI)**

Select Class: Class 2

Select Subject: Today's English Learning

| Roll | Name                  | CT1 | CT2 | HalfYearly | Yearly | Total | Grade | Edit |
|------|-----------------------|-----|-----|------------|--------|-------|-------|------|
| 9    | Ashik Hossain         | 9   | 8   | 27         | 45     | 89    | A     | Edit |
| 27   | Punit Soni            | 7   | 9   | 22         | 48     | 86    | A     | Edit |
| 7    | Adnan Sayem Chowdhury | 8   | 8   | 27         | 43     | 91    | A+    | Edit |

Submit Results

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**Figure 3.25: (SubjectWise Result Submission)**



| Roll | Name                  | CT1 | CT2 | HalfYearly | Yearly | Total | Grade | Edit              |
|------|-----------------------|-----|-----|------------|--------|-------|-------|-------------------|
| 9    | Ashik Hossain         | 8   | 8   | 24         | 50     | 41    | E     | <a href="#">✎</a> |
| 27   | Punit Sori            | 10  | 10  | 30         | 50     | 0     |       | <a href="#">✎</a> |
| 7    | Adnan Sayem Chowdhury | 10  | 10  | 30         | 50     | 0     |       | <a href="#">✎</a> |

**Figure 3.26:** Result Submission Field space

| Subject Name             | Total Number | Grade |
|--------------------------|--------------|-------|
| Today's English Learning | 80           | A     |
| Mathematics & Geometry   | 90.5         | A+    |
| গণিত ও জ্যামিতি          | 94           | A     |

**Figure 3.27:** ClassTeacher can view the results of specific class's

## 3.2 Detailed Methodology and Design

EduNexus was designed and developed in a systematic and well-structured way. Several architectural alternatives, database choices, and development models were considered and the most appropriate solution identified to use in the school management system. This part provides a description of the methodology, other methods that were discussed, and the rationale of the selected design.

### 3.2.1 Alternate Solutions The following solutions have been thought of

- **System Architecture:**
  1. **Alternative 1:** Desktop-Based Application (Java / C+ SQL Server).
    - a)**Pros:** Works offline, opposite enough to apply in one school.
    - b)**Cons:** Somehow hard to keep multiple systems in sync, limited remote access, parent, teacher, student portal is not the best.
  2. **Alternative 2:** Web-Based Application (React + Node. js + MongoDB).
    - a)**Pros:** Available everywhere (teachers, parents, students), multischool/branchable, centralized data storage, less difficult to maintain update, leverages real-time attendance and update of results.
    - b)**Cons:** Internet connectivity is needed, initial setup of hosting required.
  3. **Selection:** Web-Based Application.

The reason why it was selected is that a school management needs to have multi-user access, data stored centrally and is scalable, which a web application can provide.

- **Database Choice:**
  1. **Alternative 1:** SQL Server (Relational) / MySQL: Rigid structured tables, academic databases are popular, good when relational data are

to be used regularly.

2. **Alternative 2:** MongoDB (NoSQL): Dynamic school data (attendance, results, announcements) schema that can be easily changed, quick modern web apps development, easy to scale horizontally.
3. **Chosen:** MongoDB - Chosen as the one that is flexible in adding and modifying school-related data types quickly and is easier to integrate with technologies based on JavaScript.

- **Development Model:**

1. **Alternative 1:** Agile Methodology: Iterative development, highly flexible, needs to be collaborated with by the user continuously. It does not suit set deadlines in academics.
2. **Alternative 2:** Waterfall Model: Linear and structured, every stage passed and transformed to the next, gives clear documentation, appropriate in projects that have known needs.
3. **Chosen:** Waterfall Model: The reason is that the project requirements were well defined, the deadlines were academically fixed were and the Waterfall is appropriate since the university projects are documentation oriented.

### 3.2.2 Selected Methodology

The Waterfall Model was applied in the creation of EduNexus and it proceeds in consecutive phases: problem identification, requirement analysis, system design, implementation, testing and validation, deployment, maintenance. This model guaranteed that the whole system was developed in a well organized, predictable and well documented way.

### 3.2.3 System Design Overview

EduNexus system is developed as web based client-server application having the following components:

- **Frontend:** Web app based on React Framework and Tailwind CSS to create responsive, accessible, and user-friendly interfaces to teachers, students, admin, and parents.
- **Backend:** Express.js as a server-side logic, REST API, authentication, and role-based access control framework built on top of Node.js.
- **Database:** MongoDB to store student records, attendance records, results, fee transactions, teacher records and communication notices.
- **User Roles:** Roles at EduNexus are clear, permission based:
  1. **Admin** - controls the school activities, users, classes, charges, and system configurations.
  2. **Teacher** - deals with student attendance, student performance, classroom activity, and sharing of resources.

The system design is flexible, secure and should be scalable to maintain the real time operations in school and further upgrades.

## 3.3 Project Plan

The planning of the creation of EduNexus was done in a systematic and well-organized way so that the project can be completed within the required academic time frame. The complete project was split into several stages to allow the effective monitoring, performance, and

completion of every activity.

## **Phases of the Project Plan**

### **1. Initial Phase (Week 1-11)**

The issue of the project was chosen and determined. An extensive literature search on school management systems, ICT in education and digital transformation in academic institutions was done. The current school information systems were researched to know what they did and did not do well. In this stage the problem statement was narrowed down, project scope outlined and the initial progress report was made.

### **2. Requirement Analysis (Week 12-15)**

The difficulties in the conventional school management processes, which include manual roll call system, paper reports preparation system, fee tracking problems, and delay in communication were thoroughly examined. According to this research, the non-functional and functional requirements of the EduNexus were recorded, and the entire system requirements were completed. Module and expected output were given.

### **3. System Design (Week 16-20)**

At this stage, the whole system architecture of EduNexus was developed. The following were created:

- Data Flow Diagrams (DFD)
- Entity-relationship Diagram (ERD)
- Student, teacher, attendance, fees and results database structure.
- Admin, Teacher, Student and Parent dummy UI/UX.

This design phase gave it a proper picture of the system workflow and user interface.

### **4. Frontend Development (Week 21-28)**

The frontend was created on the React Framework and Tailwind CSS. Following modules interfaces were developed:

- Administration of students and faculty.
- Attendance management.
- Entry of results and viewing of results.
- Fee management.
- Communication tools and notice board.

The interfaces were all to be responsive, simple and untechnical to the non-technical users such as the teachers and the parents.

### **5. Backend Development and API Integration (Week 29-36)**

Backend was designed on the basis of Node.js and Express.js. REST APIs were created for:

- Role management and authentication of users.
- Submission of attendance and retrieval of attendance.
- Result management.
- Billing documents and financial management.
- Notices and announcements.

It was a MongoDB database linked with the backend. This was the stage where the frontend and the backend were combined into one system that was fully operational.

### **6. Module Implementation (Week 37-40)**

Other important modules were introduced and they included:

- Automated grade computing.

- Student progress monitoring portal.
- Admin reporting module (attendance report, fee report, performance report)
- Timetable management.

These were additions that brought EduNexus to completion.

### 7. Testing and Validation (Week 41-44)

Unit testing and integration testing were done. Testing focused on:

- Role-based access control.
- Accuracy of submission of attendance.
- Result calculations.
- Fee invoice generation.
- Communication modules.

Bugs and inconsistency were eliminated and performance optimized.

### 8. Deployment and Hosting (Week 45-46)

The build production was ready and launched into a secure cloud hosting platform. Environment variables were optimized, database connection was secured, and safe communication through the requirement of use of safe communication with the use of SSL certificate was implemented as per the standard web security guidelines [23].

### 9. Final Report and Presentation (Week 47-48)

The final project documentation was made and handed in. An actual demonstration of EduNexus was also conducted and it demonstrated its main modules which included attendance, results, fees, communication modules and role based dash boards. The last presentation marked the features, performance, and overall effects of the system on the management of the school.

## 3.4 Task Allocation

This table depicts the timeline of the principal activities in each period of the project, from week 12 to week 48.

**Table 3.4:** Task Allocation Timeline

| Tasks                                 | Weeks |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------------------------------------|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
|                                       | 12    | 14 | 16 | 18 | 20 | 22 | 24 | 26 | 28 | 30 | 32 | 34 | 36 | 38 | 40 | 42 | 44 | 46 | 48 |
| Requirement Analysis                  |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|                                       |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| System Design                         |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|                                       |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Frontend Development                  |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|                                       |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Backend Development & API Integration |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|                                       |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Module Implementation                 |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|                                       |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Testing & Validation                  |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|                                       |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |



# Chapter 4

## Implementation and Results

In this chapter, implementation details of **EduNexus: A Web-Based School Management System** are provided and the results achieved during the development and testing stage are given. It gives the environment setup, tools and technology involved, implementation process, test procedures, performance test and final results of the system as well as a comparative analysis of the system and its final outcome.

### 4.1 Environment Setup

EduNexus was designed in a well-organized, contemporary development environment so that it should be effectively implemented and easily integrated among modules of the system. The technologies and tools were as follows:

- **Frontend Development:** React.js (To create responsive, interactive, and component user interfaces)
- **Backend Development:** Node.js with Express.js (for server-side logic and REST API development and backend)
- **Database:** MongoDB (In order to store student data, teacher data, attendance, examination results, fees, and administrative data)
- **Development Tools:**
  1. Visual studio- Visual code editor for developing the front end and backend.
  2. GitHub Version control and collaborative code management. Repository Link: [https://github.com/Amrin-Sumaiya/School\\_Management\\_System.git](https://github.com/Amrin-Sumaiya/School_Management_System.git)
  3. Postman - To test API and validate it and simulate a request.
- **Hosting / Deployment:** Railway / Azure / Vercel - This is deployed to use the frontend and the backend in Render.
- **Programming Languages:** JavaScript - Javascript is used in front end and back end development.
- **Operating System:** Windows 11 - Local development, debugging and testing.
- **Additional Tools:**
  1. npm/yarn - Package management.
  2. Browser DevTools - To debug the user interface and monitor performance.

Such an arrangement provided a smooth interface between the UI, server, and database and provided scalability and maintenance of the system.

## 4.2 Testing and Evaluation / Performance / Comparative Analysis.

### 4.2.1 Testing Methodology

To achieve reliability EduNexus was put to test by using a series of structured testing methods:

**Table 4.1:** Testing Methodology

| Testing Type | Unit Testing                                                                                                                                 | Integration Testing                                                                                                                                    | System Testing                                                                                                        | Usability Testing                                                                             |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Performed    | ✓                                                                                                                                            | ✓                                                                                                                                                      | ✓                                                                                                                     | ✓                                                                                             |
| Description  | Individual modules tested: login authentication, attendance submission, result generation, teacher enrollment, admission record calculation. | Checked interactions between frontend forms, backend APIs, and databases. Example: attendance upload → API validation → MongoDB update → confirmation. | Tested full system in realistic school conditions: concurrent attendance, result processing, admin report generation. | Conducted with teachers and admin to assess ease of use, interface clarity, and productivity. |

### 4.2.2 Evaluation Metrics

To measure EduNexus, the following were the metrics:

- **Accuracy:** Ensured proper storage and retrieval student data, attendance records, results and yearly admission records.
- **Response Time:** The amount of time to load dashboards, make attendance, create results, etc.
- **Resource Utilization:** Monitored CPU and memory usage when activities are at a peak.
- **Error Rate:** The number of failed transactions or wrong operations.
- **User Satisfaction:** Feedback gathered on teachers and the administrative team.

### 4.2.3 Performance Analysis

- **Mean response time on system:** 250-350 ms on basic functions like attendance updates, student , classes, subjects fetching , and querying of results.
- **Database optimization:** Indexing cut down query time by approximately 30 percent than in the first implementation.
- **Simultaneous use:** The system was able to sustain 50+ concurrent users (teachers, students).
- **Error rate:** Under 2, primarily caused by invalid form entry. Outcomes of the

performance tests indicate that the system is efficient and can be used in healthy day to day running of schools.

#### 4.2.4 Comparative Analysis

EduNexus was compared with the current manual system as well as some of the commonly used school management and academic automation systems to determine the effectiveness of the proposed system. The features that were compared included those features that are needed in a modern educational institution, which include student information management, attendance, result processing, fee tracking, and role-based access.

- **Manual System:** Paper based registers and spreadsheets that are in use by schools. Such ways are subject to errors by human beings, time consuming and are not updated in real time. Activities such as attendance tracking, preparation of results and collection of fees consume much manual labor to cause inefficiencies.
- **ClassTune:** The commercial school management platform is a popular platform in Bangladesh. It provides such features as class routines, reports, and communicators but needs to pay subscription fees. The system might also lack customization facilities to small and medium institutions.
- **Google Classroom:** One of the most common international platforms of virtual classes, assignment management, and communication. Although helpful in the academic interaction, it lacks such administrative functions like fees management, student attendance logs, and the full student data management.
- **MySchool ERP:** An ERP solution in a school that offers attendance, examinations, and communications with parents. It can be costly to adopt, though, and it might need training, and sophisticated functionality might be pricier.
- **Proposed System (EduNexus):** A low cost solution that is lightweight and can be customized to fit the school management operations. It incorporates functionalities like real-time attendance, automatic calculation of results, fee tracking with due alert, teacherportals, studentportals, and parent portals and role based access control. EduNexus will be user friendly, effective, and economical to small and medium schools.

**Table 4.2:** Comparative Analysis of Existing Systems and Proposed System (EduNexus)

| Feature                        | Manual System | ClassTune | Google Classroom | MySchool ERP | Proposed System(EduNexus) |
|--------------------------------|---------------|-----------|------------------|--------------|---------------------------|
| Student Information Management | Yes           | Yes       | No               | Yes          | Yes                       |



|                                                  |              |        |           |             |                   |
|--------------------------------------------------|--------------|--------|-----------|-------------|-------------------|
| <b>Attendance Management</b>                     | Yes (manual) | Yes    | Limited   | Yes         | Yes(real-time)    |
| <b>Result Generation &amp; Grade Calculation</b> | No           | Yes    | No        | Yes         | Yes(automated)    |
| <b>Class Management &amp; SubjectList Fetch</b>  | No           | Yes    | No        | Yes         | Yes               |
| <b>Role-Based Access (Admin/Teacher)</b>         | No           | Yes    | Partially | Yes         | Yes(custom roles) |
| <b>Communication/Notice Board</b>                | No           | Yes    | Yes       | Yes         | Yes               |
| <b>Subjects Managements</b>                      | Yes(manual)  | Yes    | Yes       | Yes         | Yes               |
| <b>Cloud/Web Access</b>                          | No           | Yes    | Yes       | Yes         | Yes               |
| <b>Cost</b>                                      | Low          | Medium | Free      | Medium-High | Low(custom)       |

## 4.3 Results and Discussion

In this section, results of the EduNexus system were obtained following the implementation and an analysis of the extent to which the objectives of the project were met has been given. The findings are tabulated on functionality, performance, usability, and general discussion.

### 4.3.1 Functional Results

The EduNexus system was able to carry out all the basic operations of the school's management. The system was able to:

- Keep records on student, teacher, Subjects and class.
- Real-time attendance management, instant updates to the database.
- Produce academic output automatically, such as grade calculation as well as create reports.
- Managing handle fees such as invoice creation, due monitoring and payment changes.
- Ensure secure role-based access whereby the Admin, Teacher user could only run the authorized operations.
- Advertisements and announcements should be published using a centralized communication module.
- Prepare various reports including attendance summary, yearly admission and Academic Result performance sheets.
- These findings indicate that EduNexus meets all the critical functional specifications of a school management system in the modern world.

### 4.3.2 Performance Results

Sample academic data and simulated school operations was used to perform performance

testing.

- **Response time:** 180-310 ms on average response time on major operations like attendance submission, result retrieval, and the loading of fee history.
- **Concurrent usage:** System was able to support up to 60 simultaneous operations without any apparent lag.
- **Database optimization:** Indexing decreased search and record retrieval time of students by about 38%.
- **Error rate:** Has not surpassed 2% and has been partially because of form errors and invalid form entries.
- The findings validate the fact that EduNexus is optimized with speed, concurrency, and stability.

### 4.3.3 Usability and User Feedback

During testing, the system was tested by 10 users (teachers, administration):

- **Ease of use:** Rated 4.7/5
- **Interface clarity:** Rated 4.6/5
- **Overall satisfaction of the user:** 91 percent of users felt that digital management was better compared to manual record-keeping.
- **Time savings:** Time on attendance made a difference of 5-7 minutes (manual register) to less than 1 minute.
- The time spent preparing results decreased by a maximum of 60 percent.
- Tracking fees became immediate as opposed to ledger-based procedures.
- Customers have complimented the user-friendly interface, ease of use, and less paperwork.

### 4.3.4 Discussion

The findings substantiate the fact that EduNexus can fulfill its major goals: to automate the main processes of a school, to decrease the number of manual operations, as well as to enhance the precision and availability of school data.

EduNexus is much faster in saving time than the traditional manual processes and also eradicates the frequent errors of attendance, processing results, and tracking fees. The proposed system is an economical and simple to implement solution that is specific to small and mid-sized learning institutions as opposed to commercial alternatives.

The current version lacked some of the sophisticated features, like the ability to connect with mobile apps, AI-driven performance analytics, and in-depth behavior tracking, which can be implemented in the future releases.

## 4.4 Summary

The chapter provided the implementation, testing, and evaluation of EduNexus system. The basic functionalities like attendance control, result processing, fees control, and role access were all integrated. The outcomes of the testing include high performance, low error rates, and high satisfaction among the user.

Comparison has shown EduNexus to be much better than the traditional manual processes and a viable, low cost substitute to the current school management systems. On balance, the conclusions provide an idea that the system is rather effective in achieving its goals and that it is an effective and scalable solution to school administration.

# Chapter 5

## Engineering Standards and Design Challenges

In the chapter, the author introduces the criteria applied in the EduNexus project and the key issues that were experienced in the design and development stages. It further justifies and supports the choice of the engineering methods used and explains why the methods have been chosen to address advanced design issues in the field of educational information systems.

### 5.1 Compliance with Standards

When developing EduNexus, a number of standards of software, hardware, and communication were selectively taken to guarantee quality, interoperability, security and long term sustainability. These standards were used to direct the project lifecycle since planning up to deployment.

#### 5.1.1 Software Standards

##### 1. IEEE - Software Requirements Specification (SRS) Standard [11].

- **Alternate:** Ad hoc or informal requirements gathering.
- **Benefits:** Reduced preparation time, reduced documentation.
- **Disadvantages:** The probability of ambiguity and omission of requirements is high.
- **Rationale:** IEEE 830 was chosen to make sure that the EduNexus requirements would be complete, defined and traceable, particularly since school systems need a high degree of clarity on the part of student record, staff responsibilities, attendance tracking and examination administration.

##### 2. IEEE 1016 - Software design description standard [12].

- **Alternative:** Informal or unstructured design practices.
- **Benefits:** Saves original design time, very flexible.
- **Cons:** Not easy to scale, hard to work with multi-developers.
- **Rationale:** IEEE 1016 had an orderly method of recording system architecture, modules, data flow and interactions. This provided transparency and uniformity in development of student management, attendance, academic records and teacher workflows.

### 3. Software Quality Model ISO/IEC 25010 [13].

- **Alternate:** Functional testing.
- **Benefits:** Resource-less.
- **Disadvantages:** Ignores usability, performance, maintainability and security.
- **Rationale:** EduNexus was evaluated based on ISO/IEC 25010 on the standard qualities of usability (to teachers and other staff) and reliability (during peak school periods), performance, maintainability, and security which is necessary in dealing with sensitive student records.

## 5.1.2 Hardware Standards

### 1. IEEE 802.3 - Ethernet Standard

- **Alternate** Proprietary networking equipment.
- **Benefits:** Excellent functioning in directed conditions.
- **Cons:** Costly, less universal
- **Rationale:** Ethernet was developed and tested because it provides a reliable connection between client computer and the local server in order to access student records, attendance logs and administrative dashboards.

### 2. ARM Hardware Compatibility Standards x86.

- **Interchange:** Vendor/custom hardware.
- **Benefits:** Niche use case optimization.
- **Disadvantages:** costly and inefficient to scale.
- **Rationale:** The development and testing was done on commonly found PC/server hardware to ensure compatibility and scalability across the various school environments including low-budget computer labs.

## 5.1.3 Communication Standards

### 1. X.509 Certificates TLS/SSL Encryption [23].

- **Alternate:** HTTP without encryption.
- **Pros:** More set up friendly.
- **Disadvantages:** Endangers sensitive information of students.
- **Rationale:** TLS/SSL has been put in place to ensure security between the client interface and the server which ensures security of attendance data, personal student data, teacher accounts, and exam records.

### 2. Restful API Standards (HTTP/1.1, JSON Format) [22].

- **Synonym:** Proprietary or binary communication protocols.
- **Wins:** Could be quicker in small systems.

- **Disadvantages:** low portability and maintainability.
- **Rationale:** RESTful APIs have been selected because of its simplicity, scalability, and wide adoption in the industry and therefore it can be easily used to communicate between modules in EduNexus including attendance, exam management, and reporting modules.

### 3. IEEE 802.11 - Wi-Fi Standard

- **Alternate:** Wired connections
- **Alternate:** Wired connections Fully
- **Pros:** Extremely stationary device stable.
- **Cons:** Restricts the movements of teachers or classroom tablets.
- **Reason:** Wi-Fi facilitated a mobile application of EduNexus in laptops, smartphones and teacher devices in real time when it comes to attendance taking in classrooms or managing classroom activities.

## 5.2 Social, Environmental and Sustainability Impact.

This segment underscores the impact of EduNexus on learners, educators, communities and views of long term sustainability.

### 5.2.1 Impact on Life

EduNexus tremendously enhances the efficiency of school administration by digitizing the information on attendance, results and students. As soon as student performance records are available, teachers could implement the appropriate remedial measures, and parents could be informed by automated notifications. These enhancements help in achievement of improved learning results and lightening the load of staff administrators.

### 5.2.2 Society and Environment Effect.

**Society:** Minimizes mistakes in the administration and improves accuracy of the records. Brings about accountability among parents, teachers and school administrators. Promotes the use of online education and evidence-based school decisions.

**Environment:** Reduces paper consumption by scanning the attendance sheets, report cards and admission forms. Limits the physical storage requirements, which is a source of less waste. Promotes the sustainability of school activities in line with the paper-reduction efforts across the world.

### 5.2.3 Ethical Aspects

EduNexus deals with delicate student, parent and staff data. Ethical considerations contain:

- Safe keeping of academic and personal data.
- Securing grading and attendance records transparency.

- Implementing role-based access to prevent abuse.
- Ensuring that there is confidentiality and data integrity.
- Meeting privacy and security regulations of institutions of learning.

The system values fairness, accountability and privacy in its interactions.

### 5.2.4 Sustainability Plan

EduNexus will be sustainable both technically, economically and operationally:

- **Technical Sustainability:** Created on the basis of open-source technologies (HTML, CSS, JavaScript, PHP/MySQL), which minimizes the reliance on costly software solutions.
- **Economic Sustainability:** Schools of any size will be able to afford it and adopt it with ease without straining their budgets in the long term.
- **Operational Sustainability:** The system can be extended with the school requirements (e.g. new departments, classes, or reporting options) due to easy maintenance, regular backups, and modular design.
- **Social Sustainability:** The system enhances sustainability in the development of education within the communities by digitizing academic processes, reducing paper wastage, and increasing transparency in academic processes.

## 5.3 Project Management and Financial Analysis

This section will provide the cost estimation of the development of EduNexus and another cost structure that would be appropriate to implement the program in professional institutions in the real world. It also involves an estimated revenue model to be potential commercialization of the School Management System.

### 5.3.1 Budget Analysis

The chart below is that of the suggested budget needed to design EduNexus at the student level. Being an academic project, the expenses are reduced through personal hardware and open-source technologies.

**Table 5.1:** Proposed Project Budget (Student Level)

| Item                         | Quantity | Unit Cost (BDT) | Total (BDT) |
|------------------------------|----------|-----------------|-------------|
| Laptop/PC for Development    | 1        | 60000           | 60000       |
| Server Hosting (Cloud/Local) | 1        | 15000           | 15000       |
| Software Tools               | -        | 0               | 0           |

|                                  |   |      |       |
|----------------------------------|---|------|-------|
| (Open-source)                    |   |      |       |
| Internet & Utilities             | - | 5000 | 5000  |
| Human Resource<br>(Student Team) | - | 0    | 0     |
| Miscellaneous<br>(testing, docs) | - | 5000 | 5000  |
| <b>Total Estimated Cost</b>      |   |      | 85000 |

### 5.3.2 Alternate Budget (Professional Deployment)

A real-life implementation of EduNexus would need greater reliability, secured hosting, licensed software, and a specific development team. The table given below is the approximate budget of a scalable deployed production level.

**Table 5.2:** Alternate Budget (Scalable Deployment)

| Item                        | Quantity | Unit Cost(BDT) | Total(BDT) |
|-----------------------------|----------|----------------|------------|
| Professional Server Hosting | 1        | 50000          | 50000      |
| Licensed Tools & APIs       | -        | 25000          | 25000      |
| Human Resource (Developers) | 2        | 40000          | 80000      |
| Internet & Utilities        | -        | 12000          | 12000      |
| Miscellaneous Costs         | -        | 10000          | 10000      |
| <b>Total Estimated Cost</b> |          |                | 177000     |

#### Rationale

- The student budget (BDT 85,000) presupposes the personal devices, free development tools, and minimum cost of operation.
- The professional deployment budget (BDT 177,000) depicts the realistic costs like high-availability servers, licensed school APIs (SMS, attendance hardware integration), and paid developer workforce - which makes the system viable in a commercial setting in schools and institutions of learning.

### 5.3.3 Revenue Model

EduNexus would be affordable to small schools, but also could scale as required by larger schools. The next revenue model is the description of potential streams to earn money on the system.

**Table 5.3:** Proposed Revenue Model for EduNexus

| Revenue Stream                         | Description                                                | Rate(BDT)         |
|----------------------------------------|------------------------------------------------------------|-------------------|
| <b>Installation &amp; Training Fee</b> | One-time setup cost, admin training, initial configuration | 12000 – 18000     |
| <b>Monthly Subscription</b>            | Basic: Attendance + Student Info + Notice Board            | 1500/month        |
|                                        | Premium: + Exams, Fees, Reporting, Parent Portal           | 3500/month        |
| <b>Value-Added Services</b>            | Cloud backup & secure database storage                     | 600/month         |
|                                        | SMS alerts (attendance, exam results, fee due)             | 1 per SMS         |
|                                        | Custom school analytics & academic reports                 | 1200/month        |
| <b>Scalability (Multi-Campus)</b>      | Additional cost per new branch/campus                      | 1500/month/campus |

## 5.4 Complex Engineering Problem

This part demonstrates how **EduNexus: A Web-Based School Management System** solves Complex Engineering Problems (EP1-EP7), Knowledge Profile (K1-K8), and Complex Engineering Activities (EA1-EA5). The mappings are given in the tables below with short explanations given on the attributes.

### 5.4.1 Complex Problem Solving

**Table 5.4:** Mapping with Complex Engineering Problem.

| EP1<br>Dept of Knowledge | EP2<br>Range Of Conflicting Requirements | EP3<br>Depth of Analysis | EP4<br>Familiarity of Issues | EP5<br>Extent of Applicable Codes | EP6<br>Extent Of Stakeholder Involvement | EP7<br>Interdependence |
|--------------------------|------------------------------------------|--------------------------|------------------------------|-----------------------------------|------------------------------------------|------------------------|
| ✓                        | ✓                                        | ✓                        | ✓                            | ✓                                 | ✓                                        | ✓                      |



## Explanations

✓ **EP1 (Depth of Knowledge):** EduNexus needed to be informed of various fields like academic processes, academic attendance automation, and fee management, database architecture, cloud computing, and security measures. This required multidisciplinary knowledge in computing.

✓ **EP2 (Conflicting Requirements):** The project needed high speed availability to user login and features working flow. Admin and teachers' activities are different. Teacher access is dependent on the admin permissions.

✓ **EP3 (Depth of Analysis):** To choose the appropriate technology stack (React, Node.js, MySQL/MongoDB) and create school-specific modules (attendance, results, admission record), it was necessary to conduct profound analysis and evaluation of the options.

✓ **EP4 (Familiarity of Issues):** Computerization of school systems in Bangladesh (particularly medium-sized schools) is a developing phenomenon. This necessitated knowledge of local peculiarities like lack of digital literacy among the staff and poor infrastructure.

✓ **EP5 (Applicable Codes):** The project involved compliance to education data protection, privacy ethics, useability, software engineering guidelines, and secure communication (SSL/TLS).

✓ **EP6 (Stakeholder Involvement):** The stakeholder involvement in EduNexus is paramount since administrators, teachers, accounts will be working then will have varying expectations.

✓ **EP7 (Interdependence):** There are interconnected modules like attendance, exams, fees, student information, reports and notifications. One bad module will influence other modules, meaning that there is high interdependence of the systems.

## Mapping with Knowledge Profile

The table below maps EduNexus development process using the Knowledge Profile it requires to solve EP1-level problems.

**Table 5.5:** Mapping with knowledge Profile.

| K1<br>Natural<br>Science | K2<br>Mathematics | K3<br>Engineering<br>Fundamentals | K4<br>Specialist<br>Knowledge | K5<br>Engineering<br>Design | K6<br>Engineering<br>Practice | K7<br>Comprehension | K8<br>Research<br>Literature |
|--------------------------|-------------------|-----------------------------------|-------------------------------|-----------------------------|-------------------------------|---------------------|------------------------------|
|                          |                   | ✓                                 | ✓                             | ✓                           | ✓                             |                     | ✓                            |

## Explanations

✓ **K3 (Engineering Fundamentals):** Basic concepts of software engineering, relational

database design and structured system development were a must.

✓ **K4 (Specialist Knowledge):** Expert knowledge of how things are done at school- how attendance works, how to grade exams and how to take fees and communicate with parents.

✓ **K5 (Engineering Design):** EduNexus entailed the designing of user accounts, workflow of the system, dash boards, automated attendance, and multi-level reports.

✓ **K6 (Engineering Practice):** Real-life skills like creation of REST API, authentication, data safety, and deployment to a cloud server were used.

✓ **K8 (Research Literature):** School management systems and academic literature were examined to determine the gaps, to adopt the best practices, and to shun the old trends.

## 5.4.2 Engineering Activities

### Mapping with Complex Engineering Activities

**Table 5.6:** Mapping with Complex Engineering Activities.

| EA1<br>Range of re-<br>sources | EA2<br>Level of<br>Interaction | EA3<br>Innovation | EA4<br>Consequences<br>for society and<br>environment | EA5<br>Familiarity |
|--------------------------------|--------------------------------|-------------------|-------------------------------------------------------|--------------------|
| ✓                              | ✓                              | ✓                 | ✓                                                     | ✓                  |

#### Explanations:

✓ **EA1 (Range of Resources):** Cloud hosting services, open-source tools, school datasets, communication modules and human expertise (admin and teachers) are some examples of what the project needed in the process.

✓ **EA2 (Level of Interaction):** The top-level contact was made with the school stakeholders (teachers, administrators, students, guardians) to make sure that the system was aligned with the needs of the school.

✓ **EA3 (Innovation):** EduNexus brings about online attendance, automatic generated result slips, parent alerts, online payment and unified academic portals--novel features of most of the local schools.

✓ **EA4 (Impact on Society and Environment):** There is less paper work, more transparency, and, more crucially, less work by the administration with digital records. Education quality is increased through better communication and faster access to information.

✓ **EA5 (Familiarity):** Such issues as automated grading, safe student information, and built-in parent portals extend far beyond conventional coursework, which makes the project a complicated engineering undertaking.

## 5.5 Summary

This chapter addressed the engineering requirements, procedures and difficulties faced in the process of designing EduNexus. It has discussed software, hardware, and communication standards in addition to their substitutes and reasons. There was also assessment of societal, environmental, ethical and sustainability impacts. The chapter also introduced project management and financial analysis, which consisted of student level and professional deployment budgets and a projected revenue model. Lastly, the system has been mapped to intricate engineering issues, expertise signage, and engineering engagements to prove scholarly worth and engineering pertinence. On the whole, the system has achieved technical objectives, is in line with international standards as well as offers a long-term solution to school automation.

# Chapter 6

## Conclusion

The chapter provides the general overview of EduNexus project, limitations and possible future development recommendations.

### 6.1 Summary

**EduNexus** is based on the idea of modernizing school management through digitisation of student management, attendance register, academic performance of students, fee payment and communication among teachers, students and parents. The system offers an affordable, scalable, and easy-to-use solution based on open-source technologies, including React, Tailwind CSS, Node.js, Express, and MongoDB, which is applicable to small and medium-sized educational institutions.

The testing proved greater accuracy of the records of the attendance, increased speed of working with the results, absence of any mistakes in calculating the fees, and great improvement of the communication process, in contrast to the manual systems. Data privacy and safe operations were guaranteed by the role-based access model to all types of users which included; Admin, Teacher, Student, and Parent.

**EduNexus** is particularly aimed at institutions that have a hard time with manual records, ineffective communication, and slow academic processes, and provide an easy-to-use and inexpensive online hub to organize school processes.

### 6.2 Limitations

Despite the achievements of EduNexus on its key goals, there are still certain drawbacks:

- The existing weaknesses of EduNexus.
- Only single school usage is supported; no centralized multi branch/school control as yet.
- There are no advanced analytics modules (performance trends, attendance insights, fee forecasting).
- None of the automated notices, fee reminders, or attendance alerts via SMS/email (it needs third party implementation).
- Not yet developed mobile application version-system can only be accessed via browsers.
- The testing on the usability and performance was done on small data and few users.
- These restrictions are a sign of potential areas of improvement of system functionality and scalability.

### 6.3 Future Work

In order to increase the functionality on EduNexus, it can be enhanced in subsequent versions:

- The description of future enhancing areas.
- Multi-School Support Centralized dashboard of centralizing all schools, branches or

institutions.

- Mobile Application Development android/iOs application to be accessed in real-time by students, teachers, and parents.
- Intelligent Analytics Beautiful dashboards indicating attendance patterns, performance trends, fee analytics, et cetera.
- Automated Notifications SMS/email attendance, fee, announcements and exam timetables notification.
- AI-Based Insights AI-generated predictive analytics, behavioral notifications and individualized learning recommendations.
- Through Biometric Devices Automated attendance through fingerprint or RFID devices.
- Offline Mode Support Local caching and automatic syncing when connection to the internet is available.
- Multi-Language Support The product is now Bengali and other languages to be able to use more.

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